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QuantWorld

Quantum Hackathon

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QUANTUM COMPUTING

for a modelled decision

Why we studied quantum mechanics?

Yesterday informed us about the physical and mathematical vocabulary. We now want to understand how that vocabulary can help us work on a decision in our hackathon.

Ingredient	What we learned	Why we will use it
Quantum States	How we describe a quantum system	To represent enconded candidate decisions
Gates	How operations transform states	To construct an algorithm
Measurement	How we obtain outcomes	To retrieve candidate solutions
Hamiltonians	How operators describe energy	To connect decision costs with a quantum algorithm

Quantum computation involves preparing a state, changing it and measuring it.

IBM Basics of Quantum Information

Quantum mechanics does not specify our decision

Knowing how a quantum system behaves leaves the application question open

"A research team has three molecular candidates and resources to follow up exactly two. It has supplied scores for their potential usefulness. Candidates A and B provide overlapping information."

→

"What would we need to specify before any computer could recommend a pair?"

The quantum system has no knowledge of our research priorities. We must express the **decision and its rules**. That is why we studied mathematical modelling.

Not everything is optimization



Like we saw yesterday, there are also applications in sensing and communication, but even for quantum computing, it is important to make an abstraction on how the system works:

- What is happening?
- Why is this happening?
- How does it happen?
- What are the implications?

So, understanding the quantum principles and the mathematical model is **not just to run routines**..., but also how to evaluate

A small decision we can evaluate ourselves

The best individual items do not necessarily form the best combination. Our order A, B, C

Candidate	Individual value
A	7
B	6
C	4
Rule	Choose only pairs & Selecting A and B together incurs an overlap cost of 5.

Selection	Calculation
A & B	7 + 6 - 5 = 8
A & C	7 + 4 = 11
B & C	6 + 4 = 10

A and B have the highest individual scores. A and C form the best pair under our assumptions because the combination matters

Binary variables describe our selections

Each complete assignment represents one candidate decision. Remember the order A, B, C

$$x_i = \begin{cases} 1 & \text{candidate } i \text{ is selected} \\ 0 & \text{candidate } i \text{ is not selected} \end{cases}$$

Assignment	Interpretation
110	Select A & B
101	Select A & C
011	Select B & C
111	Select all 3

Not possible

We can now describe every possible selection precisely. We also know how to check whether it is allowed

Connect: bitstrings and quantum basis states

We can associate each decision assignment with a computational basis state.

Domain Description	Mathematical assignment	Quantum Encoding
Select A & C	x = (1,0,1)	$\|101\rangle = \|1\rangle_A \otimes \|0\rangle_B \otimes \|1\rangle_C$

Three qubits have eight computational basis states, corresponding to the eight three-bit assignments. A general state can contain amplitudes associated with those basis states.

Later, the algorithm must address feasibility through penalties or operations that preserve the constraint. Above is just a representation, we still need to **incorporate the objective**...

Our objective becomes an encoded energy

We could maximize this fuction, but we can also construct an energy function whose low values represent desirable decisions ⟶ therefore minimizing: A & C would be -11

$$V(x) = 7x_A + 6x_B + 4x_C - 5x_A x_B$$ ⟶ Our function

$$E_{\text{objective}}(x) = -V(x)$$ ⟶ Take the negative - Energy function

$$H_C|x\rangle = E(x)|x\rangle$$ ⟶ Define the objective energy fuction through the Hamiltonean...**why?**

The energy function scores our choices. The Hamiltonian lets a quantum algorithm use those scores ⟶ we provide information to the circuit

A Quantum Approximate Optimization Algorithm

Our rules must also enter the encoding

How do we represent the “exactly two” rule? Because selecting ABC would give us 12 and -12 which is greater and smaller.

We add cost when a selection violates the rule ⟶ **penalties, example P = 10**

Numbers Selected	Added penalty
0	40
1	10
2	0
3	10

$$E(x) = -V(x) + P(x_A + x_B + x_C - 2)^2$$

The cost operation gives different assignments different phase changes.
Mixing allows those phase differences to affect the probabilities of the outcomes we later measure.

A Quantum Approximate Optimization Algorithm

What is next?

We learn how to construct and run the connection we have just understood

Knowledge we bring	Knowledge on the next block
Binary decisions and objective	The QUBO formulation and its coefficient matrix
Constraints and feasibility	Penalty construction and calibration
Basis states and Hamiltonians	QUBO/Ising translation and the cost Hamiltonian
Gates, phase and measurement	QAOA cost and mixer layers
Interpreting outcomes	The classical parameter-optimization loop and evaluation

We have now connected the decision to quantum states and an encoded cost.

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